CONCEPTUAL SERIES
MongoDB

Name: .... Cohort/Batch: .... Date: .... Score: ....

MONGODB PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is MongoDB and what are its core strengths? — *Threat Model*

Q2 How does MongoDB guarantee consistency and handle transactions? — *Architecture*

Q3 What index types does MongoDB support and when do you use each? — *Configuration*

Q4 How do you design a schema or data model in MongoDB? — *Hardening*

Q5 How does MongoDB scale horizontally? — *SSO / IdP*

Q6 How do you monitor and tune slow queries in MongoDB? — *Observability*

Q7 What backup and restore strategies does MongoDB support? — *Lifecycle*

Q8 How do you handle schema migrations in MongoDB? — *Compliance*

Q9 What security features does MongoDB provide? — *Testing*

Q10 What are common anti-patterns when using MongoDB in production? — *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 MongoDB is a relational, document, key-value, graph, Mitigation
MongoDB is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A2 MongoDB provides either full ACID transactions or Primitives
MongoDB provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A3 MongoDB offers B-tree, hash, full-text, spatial, or Hardening
MongoDB offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A4 Schema design in MongoDB involves normalizing relations Eschas
Schema design in MongoDB involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A5 MongoDB scales via replication (read replicas) for Federation
MongoDB scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some MongoDB implementations handle this automatically; others require manual configuration.

A6 MongoDB provides a slow query log, explain/explain Observability
MongoDB provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A7 MongoDB supports logical backups (export SQL or Automation
MongoDB supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A8 Schema migrations in MongoDB are managed with Governance
Schema migrations in MongoDB are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A9 MongoDB supports role-based access control, encrypted Assessment
MongoDB supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A10 Common mistakes include selecting all columns instead Optimization
Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.